

Lab Report 5

EEE 3400 :Numerical Techniques Laboratory

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Answer for Exercise 3

	u_0	u_1	u_2
x	0	2	3
y	1	2	4
	y_0	y_1	y_2

$$f[u_0, u_1] =$$

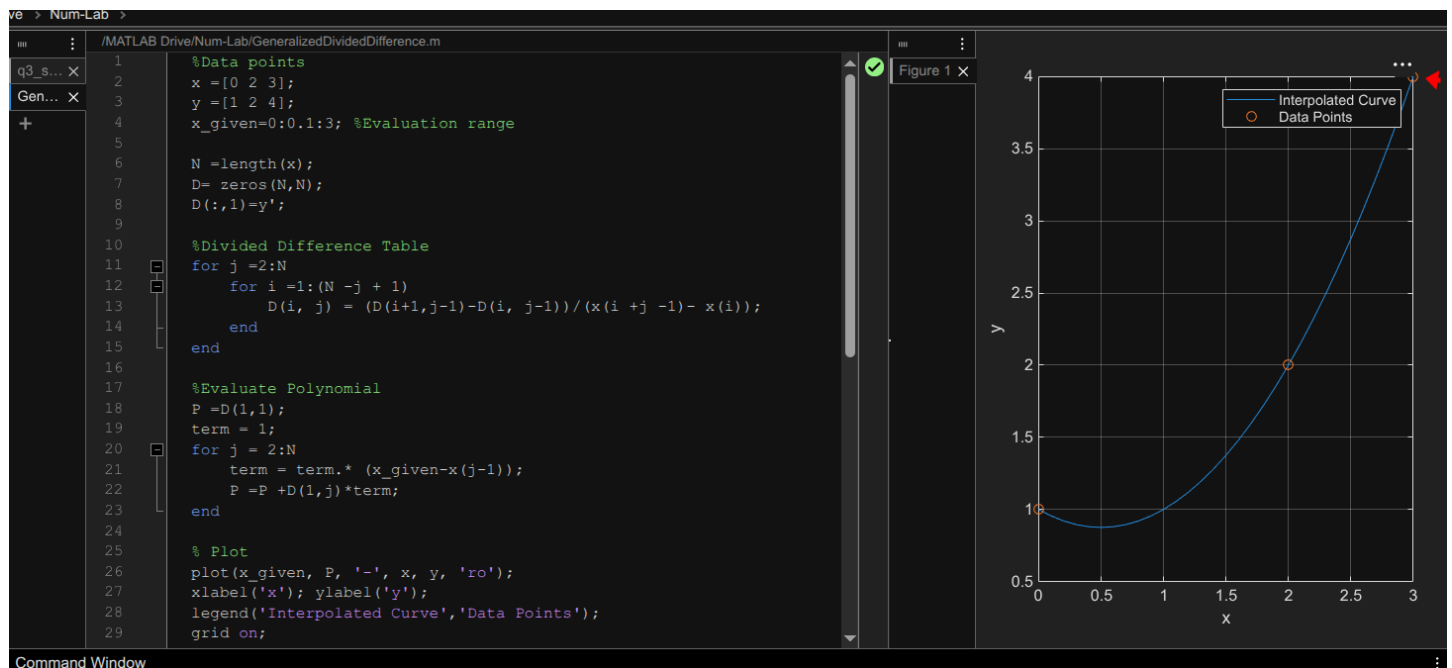
$$\left| \begin{array}{l} u_0 = 0 \\ y_0 = 1 \end{array} \right| \begin{array}{l} u_1 = 2 \\ y_1 = 2 \end{array} \rightarrow f[u_0, u_1] = \frac{2-1}{2-0} = \frac{1}{2}$$

$$\left| \begin{array}{l} u_2 = 3 \\ y_2 = 4 \end{array} \right| f[u_1, u_2] = \frac{4-2}{3-2} = 2$$

$$P[u_0, u_1, u_2] = \frac{2 - \frac{1}{2}}{3-0} = \frac{1}{2} = \frac{f(u_1, u_2) - f(u_0, u_1)}{u_2 - u_0}$$

Polynomial form $\Rightarrow P(u) = 1 + \frac{1}{2}(u-0) + \frac{1}{2}(u-0)(u-2)$

u_i	$y_i = f(u_i)$	$\frac{u_i}{1^{st} \text{ Diff}}$	$\frac{u_i}{2^{nd} \text{ Diff}}$
0	1	$\frac{1}{2}$	$\frac{1}{2}$
2	2		
3	4	2	



Matlab Code:

%data points

```
x=[0 2 3];
```

```
y=[1 2 4];
```

```
x_given=0:0.1:3; %evaluation range
```

```
N=length(x);
```

```
D= zeros(N,N);
```

```
D(:,1)=y';
```

%Divided Difference Table

```
for j =2:N
```

```
for i =1:(N -j + 1)
```

```
D(i, j) = (D(i+1,j-1)-D(i, j-1))/(x(i +j -1)- x(i));
```

```
end
```

```
end
```

%Evaluate Polynomial

```
P =D(1,1);
```

```
term = 1;
```

```
for j = 2:N
```

```
term = term.* (x_given-x(j-1));
```

```
P =P +D(1,j)*term;
```

```
end
```

% Plot

```
plot(x_given, P, '-', x, y, 'ro');
```

```
xlabel('x'); ylabel('y');
```

```
legend('Interpolated Curve','Data Points');
```

```
grid on;
```